



## CERa-MKII – Gaseous Diffusion Coefficient Apparatus

CERa-MKII apparatus for investigation of mass transfer and gaseous diffusion has traditionally used a capillary tube in a hot water bath, with a travelling microscope used to measure the rate of diffusion over a period of time. Although capable of giving good results, this type of apparatus was not without its disadvantages, particularly ease of use.

Armfield have yet again introduced a radical alternative to the traditional methods, utilising the benefits of modern computer compatible microscopy, producing a novel system with a large number of benefits.

- **The MKII version Features Electronic USB Microscope and auto sample time lapse imagery and Video**

### PRODUCT CODE: **CERA-MKII**

**Categories:** CE Series - Chemical Engineering, CERa-MKII - Gaseous Diffusion Coefficient Apparatus, Educational

**Tags:** Bench Mounted , Capillary Tube , CE Series , CERa-MKII , Chemical Engineering Basic Process Principles , Computer Compatible Microscopy , Concentration Differences , Convective Effects , Diffusion Coefficient , Diffusion Coefficients , Gas By Evaporation , Gaseous Diffusion , Gaseous Diffusion Coefficient Apparatus , Laboratory , Liquid Surface , Mass Transfer , Microscope , Presence Of A Stationary Gas , Rate Of Diffusion , Vapour Diffuses , Vapour In Air

## DESCRIPTION

CERa-MKII apparatus for investigation of mass transfer and gaseous diffusion has traditionally used a capillary tube in a water bath, with a travelling microscope used to measure the rate of diffusion over a period of time. Although capable of

giving good results, this type of apparatus was not without its disadvantages, particularly ease of use.

Armfield have yet again introduced a radical alternative to the traditional methods, utilising the benefits of modern computer compatible microscopy, producing a novel system with a large number of benefits.

A small quantity of the volatile liquid to be investigated is placed in a capillary tube, which is positioned in a heated metal block. The block is heated by an electric cartridge heater and kept at a constant temperature using an electronic control loop.

A stream of air is passed across the top of the capillary tube, with a stationary layer of air above the liquid surface, creating a partial pressure difference between the liquid surface and the flowing air stream. As vapour diffuses from the liquid into the air stream, the amount of liquid in the tube reduces and the phase boundary between the liquid and the air drops. Measurement of how this phase boundary changes with time enables the molar mass transfer rate to be determined.

A slot in the heater block enables the phase boundary to be observed using a high definition microscope. This microscope produces high definition images onto a standard windows computer (not supplied). The software provided with this microscope includes highly accurate measurement and an automated time lapse facility (either as video or multiple still images).

## TECHNICAL SPECIFICATIONS

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- A self-contained bench mounted apparatus for the determination of diffusion coefficients of a vapour in air
- The apparatus blows air across the top of the capillary tube, inside diameter 2mm
- The tube is contained in a heated aluminium block with a 25W heater
- The temperature control system maintains the block to be better than  $\pm 1^{\circ}\text{C}$  from ambient to  $60^{\circ}\text{C}$ .
- Operating temperature typically achieved in 20 minutes ( $40^{\circ}\text{C}$  operation)
- The equipment incorporates a high definition USB microscope with a resolution of  $1600 \times 1200$  and preset to an optical gain of approximately  $\times 37$
- Microscope and heater block containing the capillary are visible to the user and internal illumination of the capillary is provided
- Viewing length of the capillary is 12mm
- Physical reference scale provided (can be used for software calibration)
- Phase boundary measurement resolution  $< 0.0002\text{mm}$
- Powered by universal power adaptor with worldwide approvals
- Software can create individual images under operator control, multiple timed images under automatic control or time lapsed video. Images can be time and date tagged. Time lapse periods can be set from five seconds to many days.
- Output formats include .jpg .bmp .avi

## FEATURES & BENEFITS

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- New approach using a computer linked high definition microscope
- Very easy to see the phase boundary and to record images for analysis and reports
- Software includes accurate measurement facility for determining the rate of diffusion
- High measurement accuracy and resolution enables results to be obtained over a shorter timescale
- Software includes 'time lapse' facility to enable measurements to be taken automatically over extended time periods
- Suitable for other liquids with lower diffusion coefficients than the standard acetone (propanone) due to the 'time lapse' facility
- No water bath, small heater block is quick to set up and achieve a stable temperature
- Unlike water baths, it is safe to be left unattended for extended periods

## Experimental content

- To determine the diffusion coefficient of a gas by evaporation from a liquid surface
- Direct measurement of mass transfer rates in the absence of convective effects



- Use of gas laws to calculate concentration differences in terms of partial pressures
- Use of Fick's law to measure diffusion coefficients in the presence of a stationary gas
- Measurement of the effect of temperature on diffusion coefficients
- Investigation into diffusion coefficients of alternative fluids
- Gaining familiarity with the use of laboratory instruments to achieve accurate measurements of data required for industrial process design

## Requirements

### Electrical supply:

Requires 24V DC at 2.5A

The equipment is supplied with a universal mains adaptor suitable for 100V to 240V AC, 50/60Hz

Software requires the user to have a PC running Windows 7 or above with a USB port.

## Shipping Specifications

### PACKED AND CRATED SHIPPING SPECIFICATIONS

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**Volume:** 0.15m<sup>3</sup>

**Gross Weight:** 5kg

### Dimensions

### DIMENSIONS TRAY

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**Length:** 0.27m

**Width:** 0.20m

**Height:** 0.13m

### Order Codes

CERa-MKII

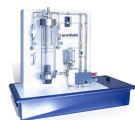
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